

Mandarin: A Culture Spanning Three Thousand Years

A Digital Humanities Approach to Cultural Semantic Evolution

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Research Background

– Background

The **mandarin** native to southern China. It is more than just a sweet fruit; it carries a cultural memory in Chinese History from Ancient Times to the Present

– Research Gap

Past research often focused on the text of poem such as Qu Yuan's "Ode to the Mandarin") for close reading, resulting in a fragmented perspective. There is a lack of macro-level, quantitative analysis spanning the 3,000-year chinese history.

– Our Approach

Using Digital Humanities Methodology (NLP & Semantic Analysis) to trace the semantic shift of the Mandarin across 8 historical eras.



Thesis Statement (Evolving Significance)

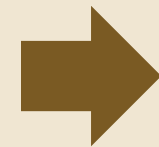
– Research Question

How has the semantic meaning of the 'Mandarin' evolved over 3,000 years?

– Path of semantic evolution of Mandarin

Phase 1: Virtue (德)

Pre-Qin Era: Symbol of Virtue
& The "Junzi" (Nobleman)



Phase 2: Tribute (贡)

Han-Tang Era: Symbol of Royal Favor,
Tribute System, & Auspiciousness.



Phase 3: Medicine (药)

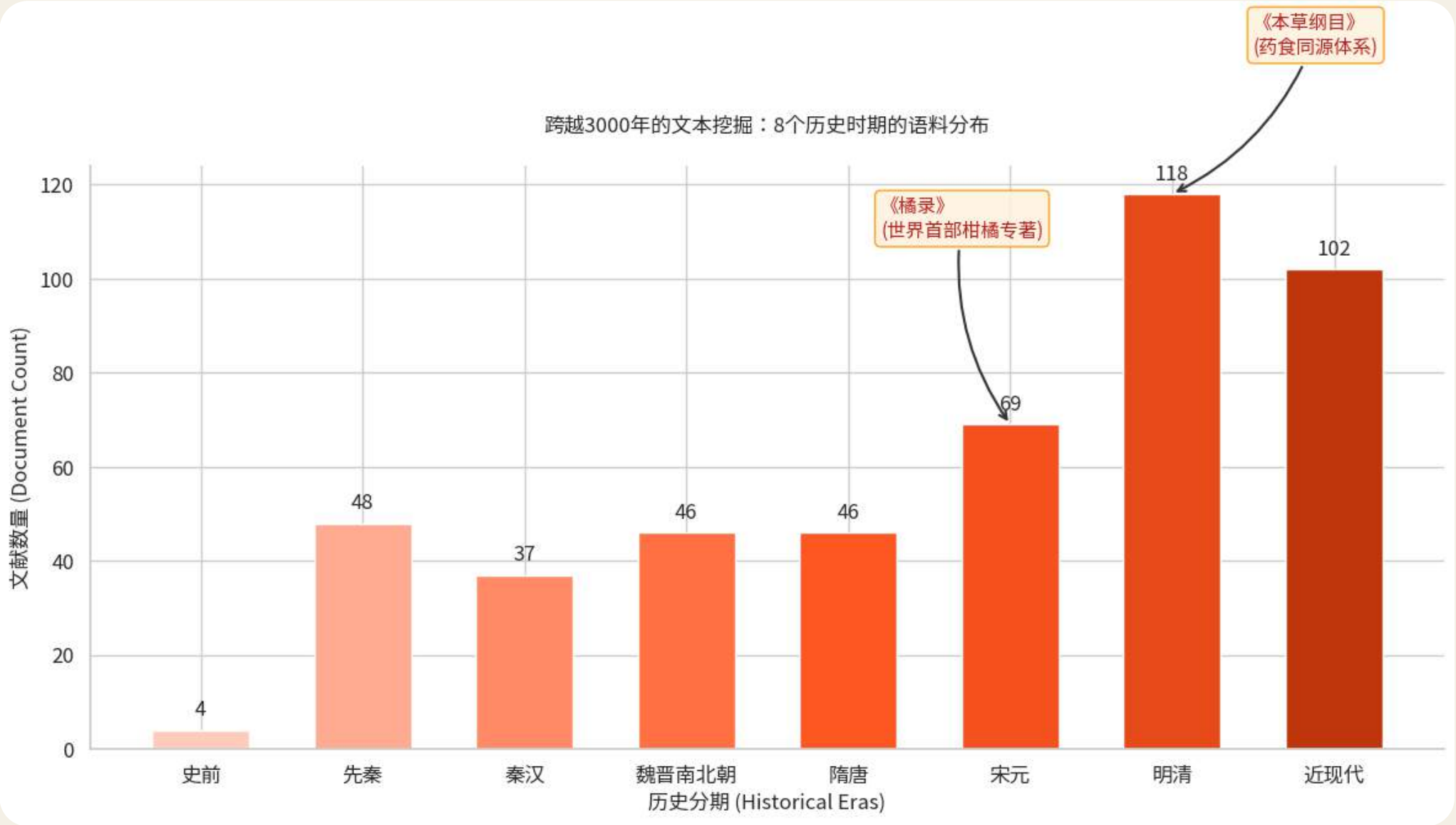
Song-Ming-Qing Era: Medicine
(Chenpi), Trade, & Practical Value.



Data & Corpus: A 3,000-Year Textual Excavation

– The Corpus: A 3,000-Year Archive

- **Data Integrity:**
A rigorous corpus of 470 distinct records.
Through cleaning process resulting in the structured dataset:
tangerine_corpus_clean.xlsx.
- **Timeline:**
Spanning 8 Historical Eras, from the Prehistoric age to the Modern era.
- **Core Sources:**
Literary: Book of History (Shangshu), Chu Ci (Ode to the Orange).
Medical: Ben Cao Gang Mu (Compendium of Materia Medica).
Agricultural: Ju Lu (The first citrus monograph).
Modern: Contemporary corpus & Industry history texts.



The Digital Humanities Pipeline

Step 1: Data Cleaning (数据清洗)

- Action: Era segmentation (分时代清洗) & Standardization.
- Goal: Transforming unstructured ancient texts into structured datasets based on 8 historical eras.

Step 2: NLP Processing (自然语言处理)

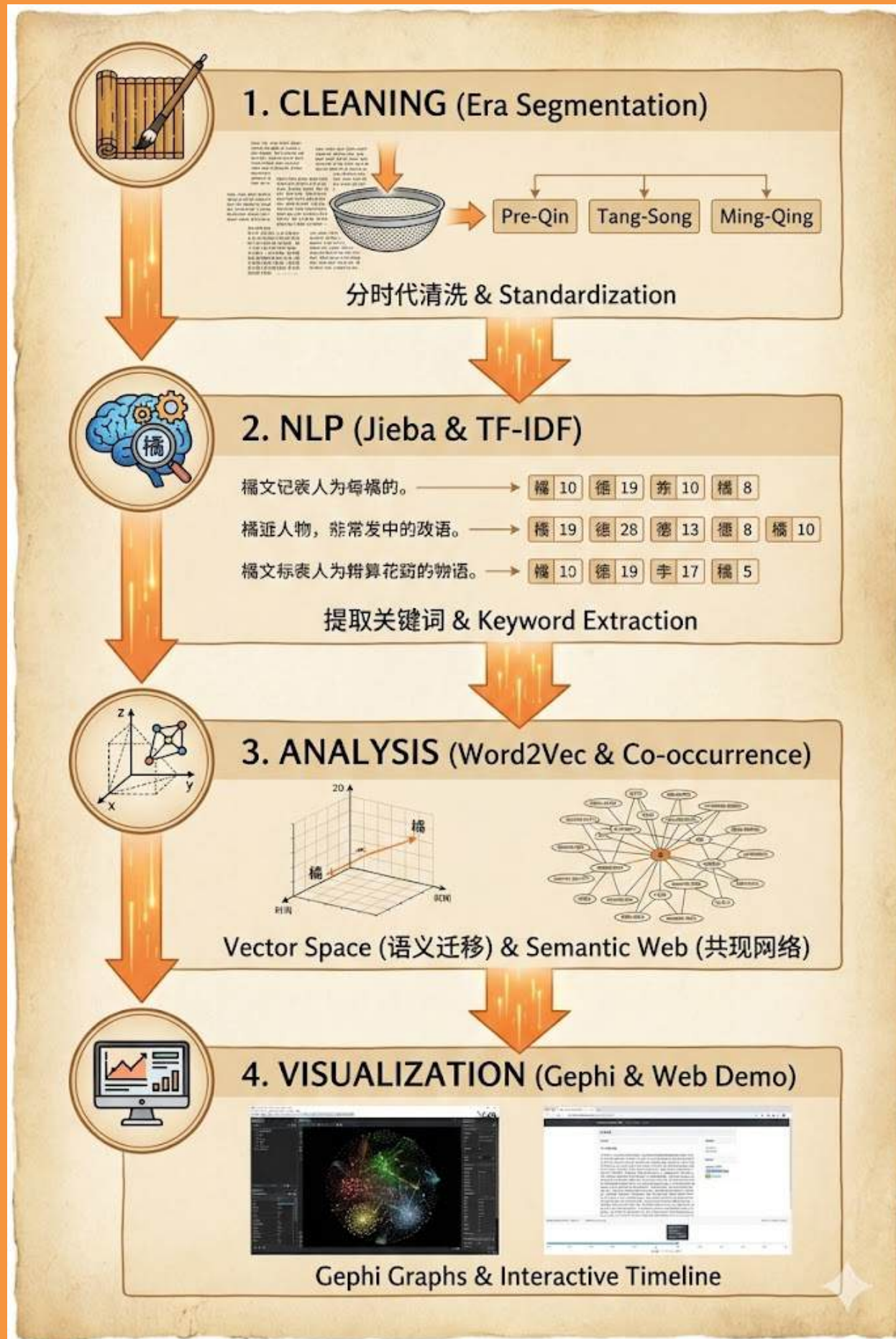
- Tools: Jieba Tokenization (结巴分词) & TF-IDF Algorithms.
- Goal: Extracting core cultural keywords and filtering out stop words to identify semantic kernels.

Step 3: Computational Analysis (计算分析)

- Word2Vec: Mapping keywords into Vector Space to measure semantic drift.
- Co-occurrence Matrix: Constructing semantic webs to reveal hidden connections between concepts.

Step 4: Visualization (可视化)

- WordCloud, Word Rank Charts, high-quality network graphs, Geographic migration map
- Web Visualization: Building an interactive timeline and dynamic graphs for user exploration.



Website demonstration

Demonstrate our DH research platform

以文化为脉络的跨时代数字人文研究

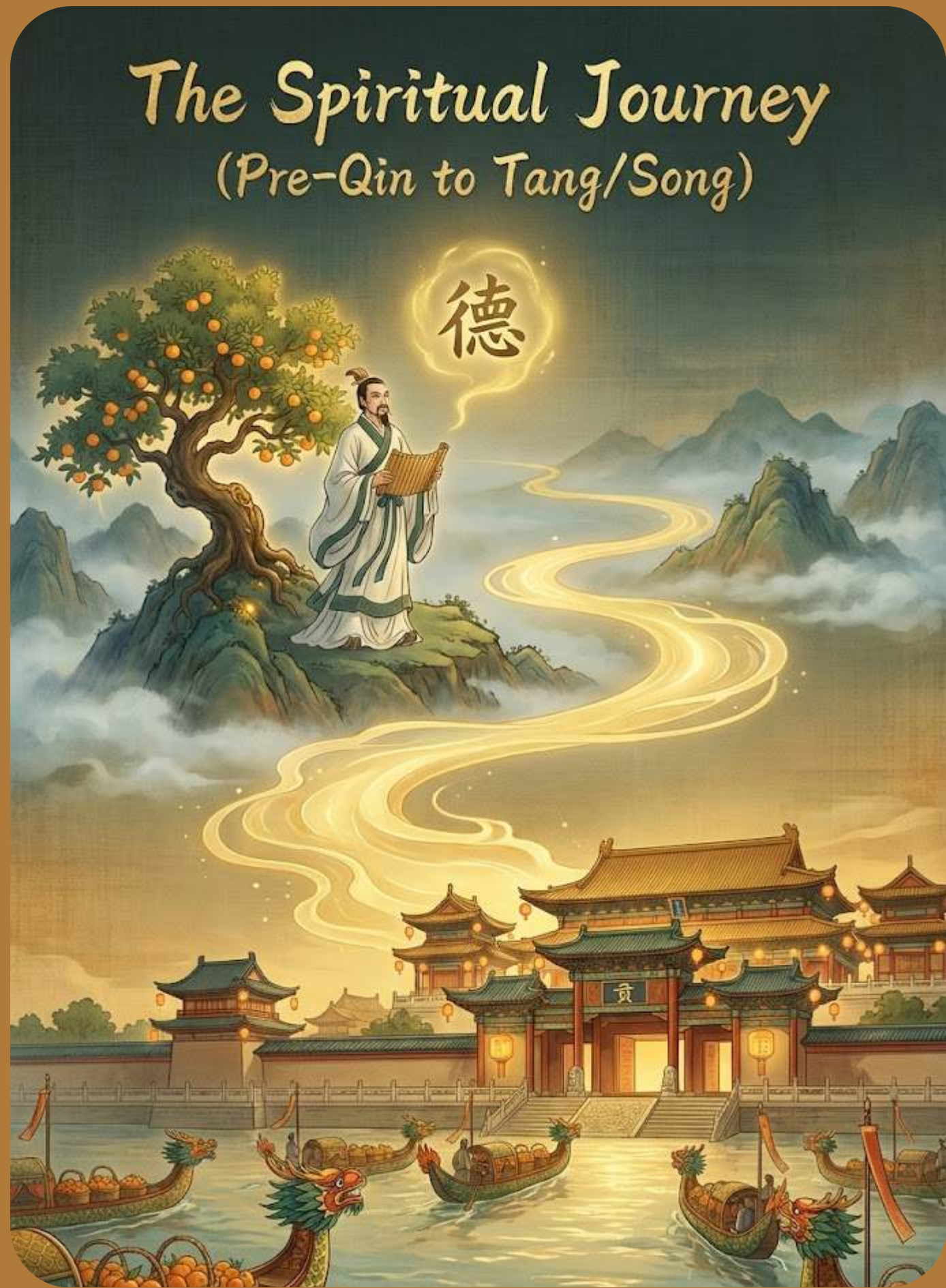
《橘：跨越三千年的文化演化》

The Mandarin: A Cultural Evolution Spanning Three Thousand Years in China

● 史前与上古 — ● 先秦 — ● 秦汉 — ● 魏晋南北朝 — ● 隋唐 — ● 宋元 — ● 明清 — ● 近现代

进入研究 / Explore the Study





The Spiritual Journey (Pre-Qin to Han)

Era 1: 史前 - 先秦 (The Symbol of Virtue)

Key Insight

- In the Pre-Qin era, the Mandarin functioned as a **moral symbol embodying Confucian values.**

Keyword Cloud Results

- High-frequency about “**virtue (忠贞),**” “**gentleman (君子),**” and “**independence (独立).**”
- These keywords show that Mandarin fruit was closely associated with **moral character** rather than material value.



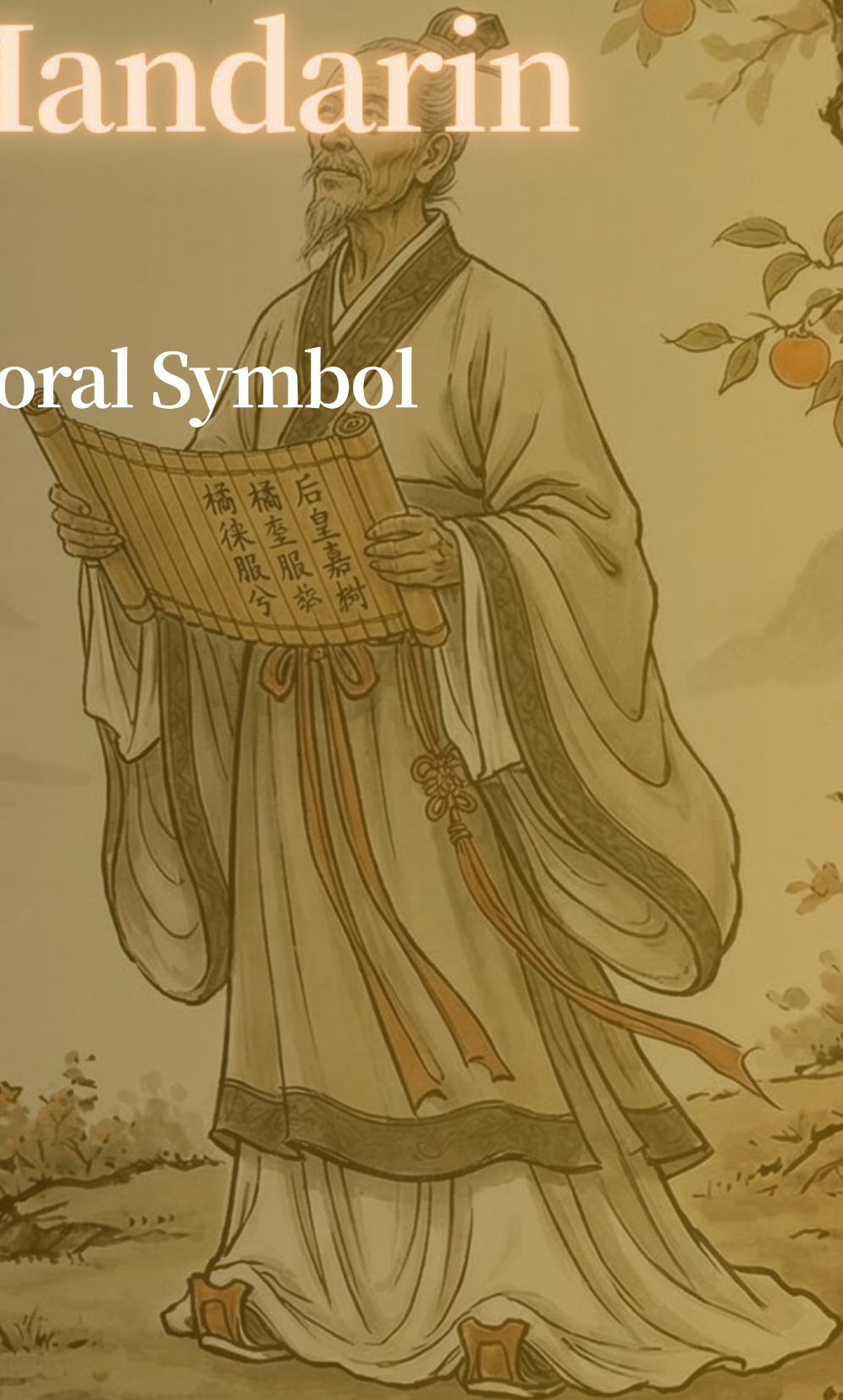
人格化的开端： 屈原与《橘颂》



Qu Yuan and the Moral Personification of the Mandarin

Qu Yuan 《橘颂》 — The Birth of Mandarin Moral Symbol

后皇嘉树，橘徕服兮。
受命不迁，生南国兮。
深固难徙，更壹志兮。



Qu Yuan and the Moral Personification of the Mandarin

Qu Yuan uses the Mandarin as a metaphor for himself:

- Unyielding, loyal, and unwilling to compromise his principles.

Cultural Significance:

- **Moral Personification:**
 - Mandarin becomes a symbol of moral integrity, loyalty, righteousness, and steadfastness
- **Long-term Literary Influence:**
 - Established the Chinese literary tradition of using the Mandarin to symbolize noble character, which continued for centuries



🍊 Analysis and Insight

Mandarin as a Symbol of Filial Piety

Cultural Meaning:

- Mandarin becomes a symbol of filial love, longing, and emotional connection between mother and child.
- The story was written into the Twenty-Four Filial Exemplars

《后汉书·梁鸿传》注
晋潘岳少时，常怀橘以遗其母。入京师，见橘而思亲，怀之以归。其母见橘而泣曰：“吾儿至孝，使我见此橘，如见吾儿。”



Mandarin as the Earliest Institutional Record

Cultural Significance:

- Mandarin was recognized as an important **tribute fruit**, symbolizing ritual order, state authority, and sacred offerings.
- Mandarin evolved from an **agricultural product** to a **state-level fiscal resource** and eventually a **ritual symbol of purity, prosperity, and political order**

凡祭祀，共其橘柚。
——《周礼》

江苏、安徽、江西、湖南、湖北等地生产的柑桔，已列为贡税之物。——《禹贡》



🍊 Analysis and Insight

Mandarin as the Earliest future-telling Record

Baoshan Chu Bamboo Slips 包山楚简的卜辞:

- “橘”keywords appears 17 times
- Record notes: “占得橘，利涉大川”

Earliest realistic Mandarin image found in China

- Unearthed a finely painted **lacquer box** from Chu culture
- The pattern depicts a Mandarin fruit —— The Earliest realistic Mandarin image found in China

Cultural Significance

- **Belief & Rituals:** 橘=Symbol of blessing and protection
- **Social Meaning:** Marker of status and spiritual value of lucky
- **Social:** Holds status, aesthetic, and symbolic value



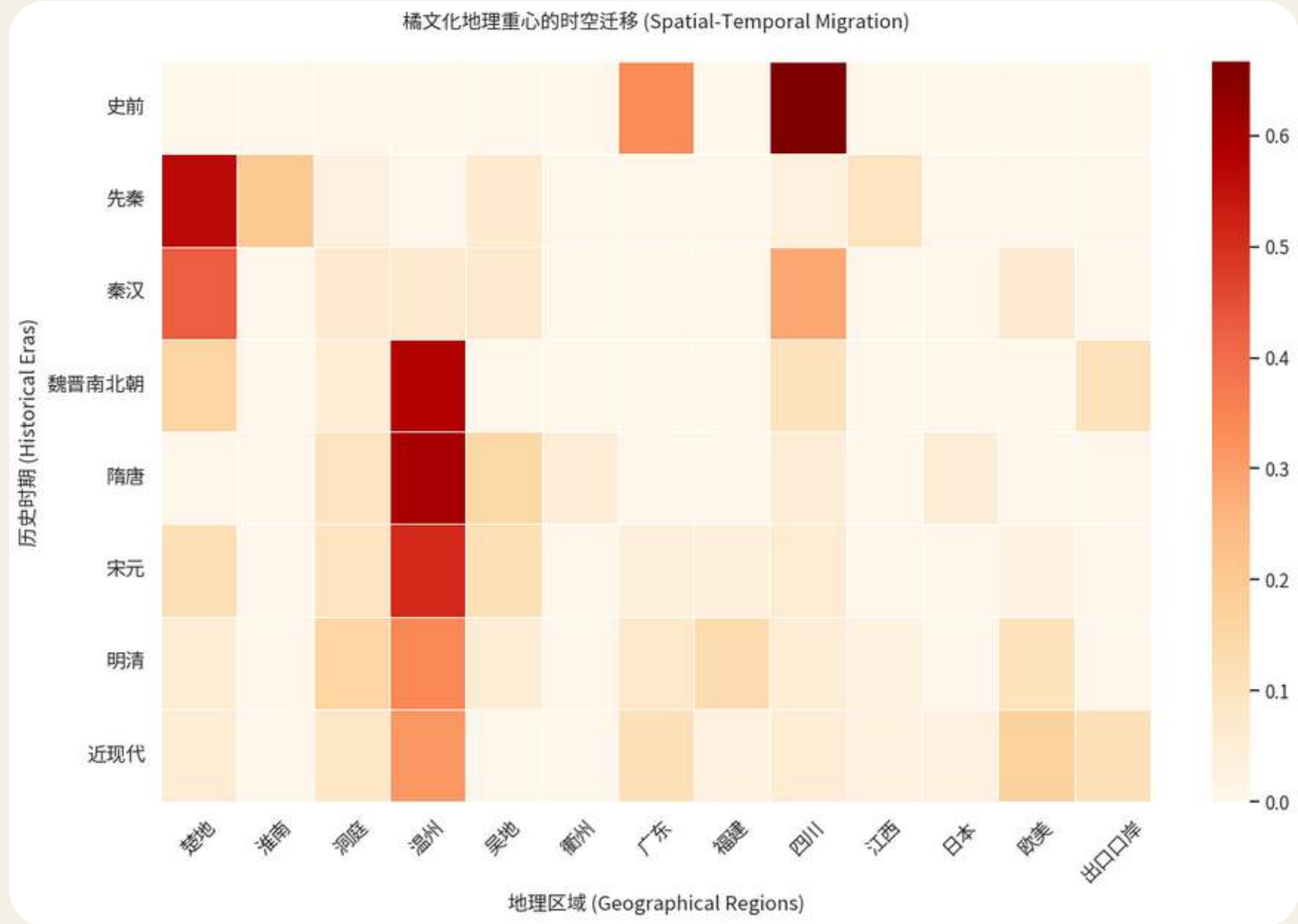
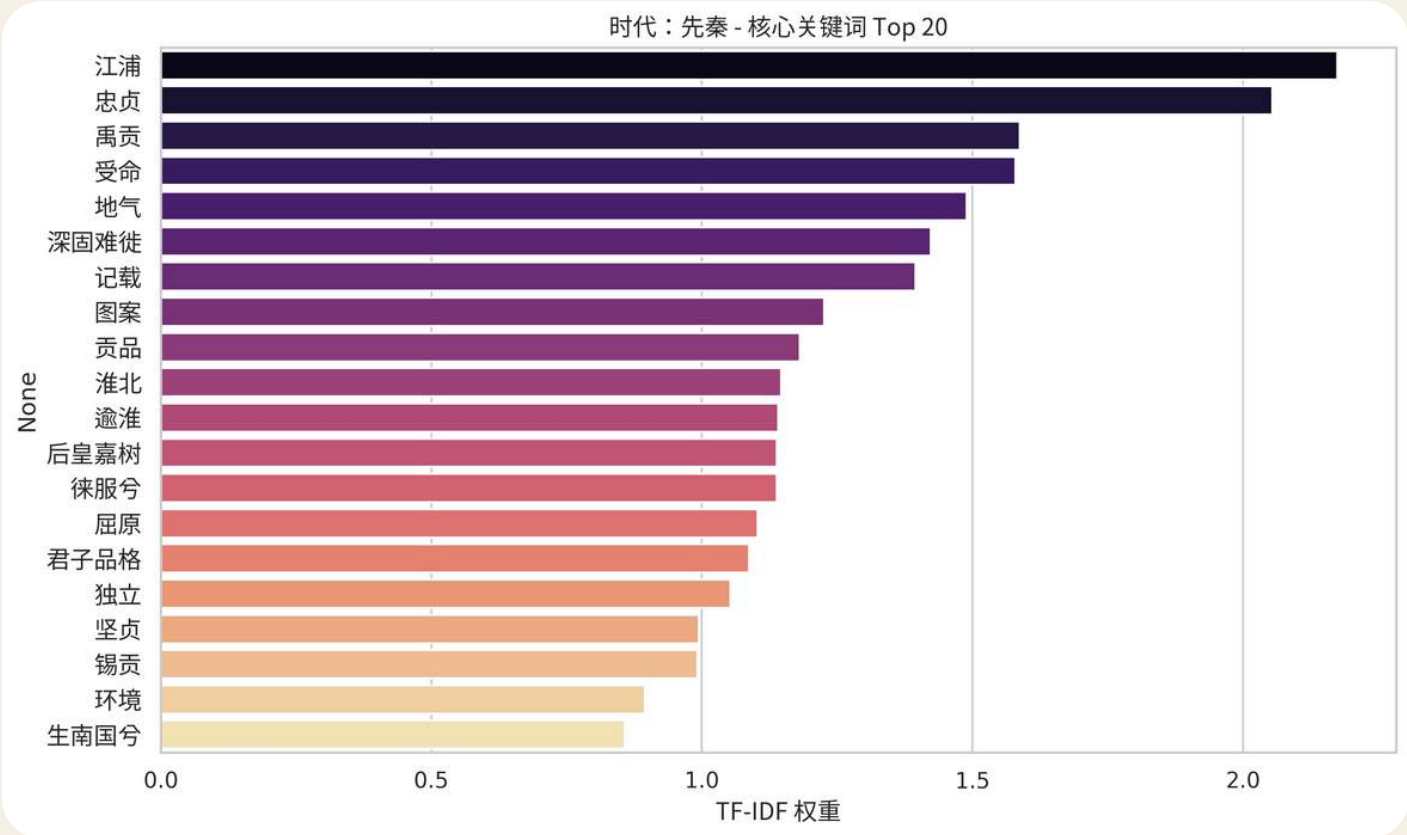
TF-IDF Analysis and Insight

Why Does “Jiangpu”江浦 Score So High?

- **High Uniqueness:**
- “Jiangpu” is strongly tied to the literary style of Chu Ci楚辞.
- **Algorithm Logic:**
- In contrast, “Jiangpu” appears almost exclusively in a Pre-Qin/Chu cultural context, making it the most representative keyword of that era.

Finding

- Initially, we expected 'Virtue' (忠贞) to be the top keyword. However, our algorithm picked 'Jiangpu' as No.1.
- It reveals the Geographical DNA of the Mandarin.
- In the Pre-Qin era, the Mandarin was strictly bound to the Southern Waters (Chu Culture).
- 起初我们以为是“德”，但算法选了“江浦”。这反而揭示了橘子的“地理基因”——它在先秦是严格绑定在南方水土的，象征着“受命不迁”的忠诚。



Mandarin Turns into Trifoliate Orange — Geography as Destiny

《晏子春秋》一橘生淮南则为橘，生于淮北则为枳。叶徒相似，其实味不同。



Cultural Interpretation

- Environment shapes nature—whether in humans, governance, culture, or morality.

Symbolic Meaning

- Human nature is not fixed; behavior changes with environment.
- Institutions and governance determine whether people flourish or deteriorate.
- Geography and culture influence the true nature of things.

Key Insight:

- The Mandarin becomes a metaphor for **how external conditions transform inner qualities**, making it one of the earliest linguistic symbols of environmental determinism in Chinese thought.





Mandarins at the Peak of Imperial Power (Han to Tang)

Mandarin as a Political Gift

By the Three Kingdoms period, Ou mandarins were already famous and prized as elite regional specialties.

三国时即被孙权选为名珍馈赠曹操，有‘果中皇帝’之誉。”

Cultural Significance:

- Political Diplomacy: Mandarins as strategic political gifts, as goodwill, alliance-building, and soft power between rival states
- Economic Significance: Their value reflects the established specialty-fruit economy of the Jiangnan region, early branding of local produce as luxury commodities.

Mandarins act as:

- tokens of political legitimacy
- markers of elite taste
- objects charged with supernatural aura, bridging material value and metaphysical imagination



Mandarin as Imperial Tribute

Wenzhou Mandarin (Ougan) as Official Tribute

- 《新唐书·地理志》载：“土贡：布、柑、橘、蔗、蛟革。”
- Their round shape and bright color symbolized:
 - Reunion
 - Light & prosperity
 - Good fortune
 - Wordplay: mandarin (柑) ≈ official (官) → symbol of promotion & success



Nationwide Tribute System: Large-Scale Tribute (7th–13th Century)

Regions submitting mandarins to the court included:

- Sichuan, Hunan, Guangdong, Guangxi, Fujian, Zhejiang, Jiangsu

Cultural Significance:

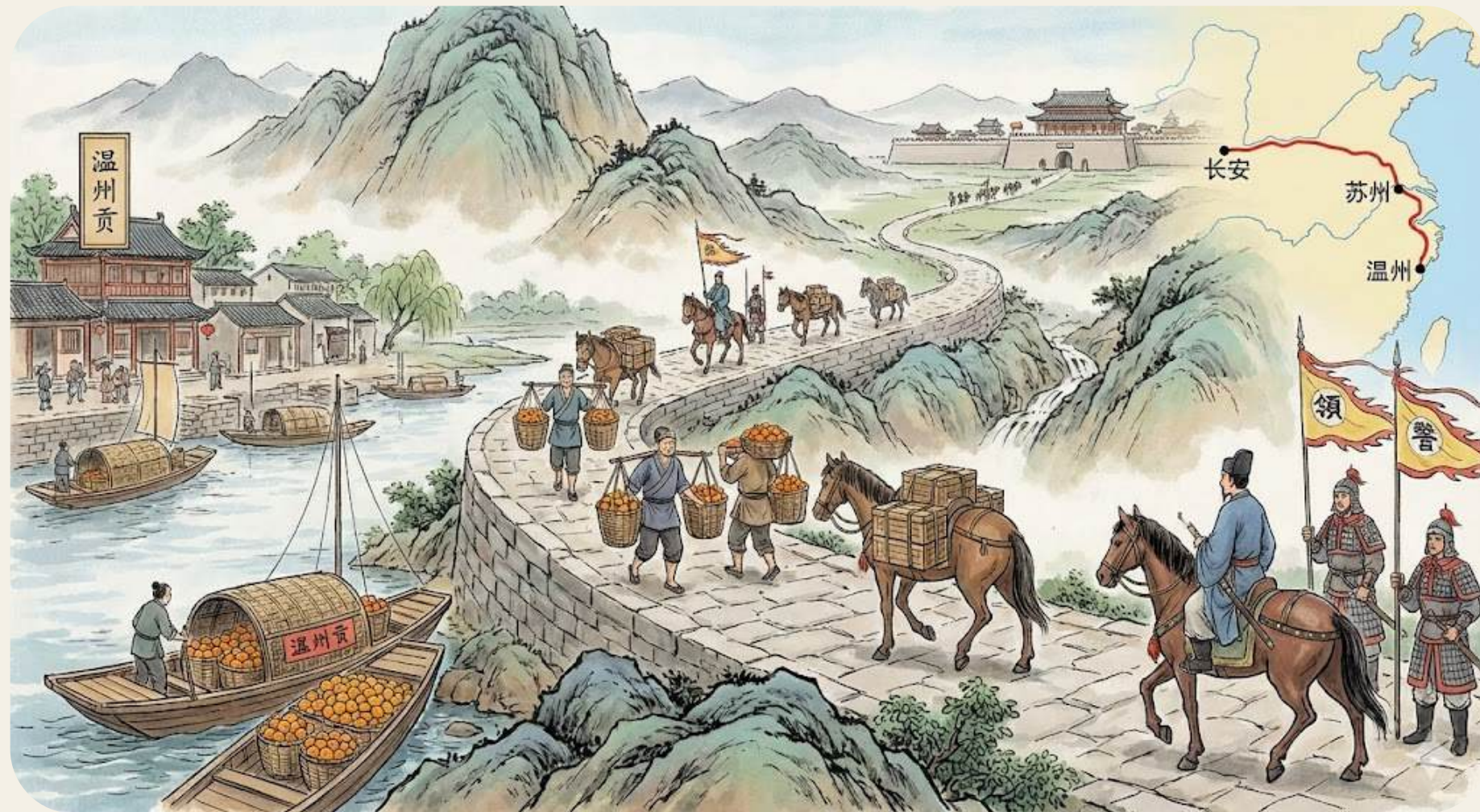
- Mandarin was a major national cash crop
- It held a stable place in the imperial tribute system



Geography: The Opening of the Southern Tribute Routes

The Geographic Imprint of Citrus Logistics

Similar to the famous "一骑红尘" for lychees, the need to transport fresh mandarin spurred the development of an efficient tribute route network, clearly outlining the geography of imperial resource flow.



杜牧《过华清宫》：“一骑红尘妃子笑，无人知是荔枝来。”

虽咏荔枝，但柑橘贡运的急迫与网络与之类似，形成了从江南核心产区直达两京的物流干线

Why were Tribute Routes Needed?

The Geography of Preservation and Power

- Mandarin is perishable, demanding high-speed transport and robust logistics, which in turn drove the maintenance and management of transportation infrastructure.
- The existence and efficient operation of these mandarin demonstrated the empire's ability to concentrate scarce resources from peripheral regions at the center of power.

Practical: Exploration of Garden Cultivation and Storage

Early Steps in Domestication and Technology

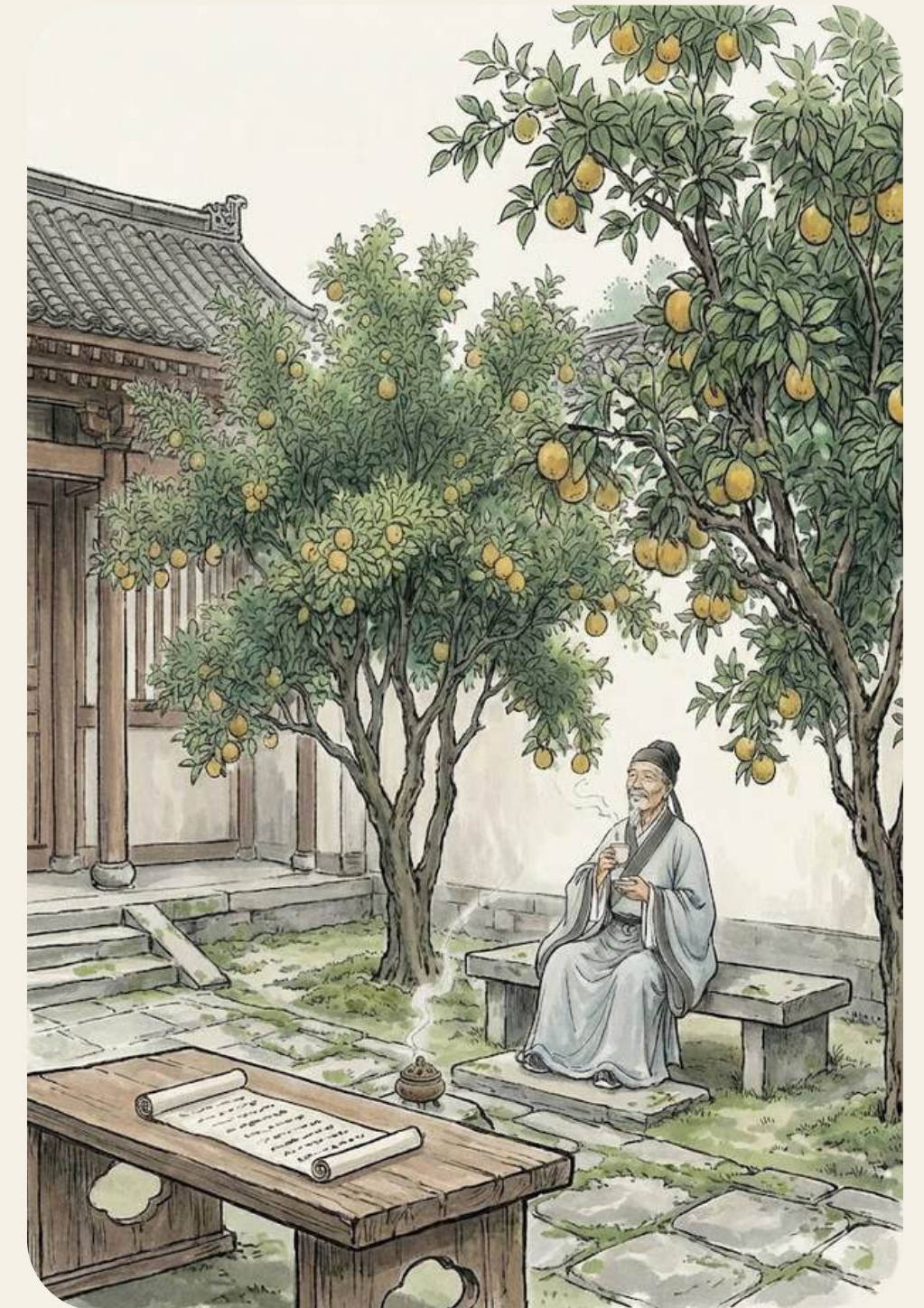
During the Tang, mandarin cultivation was no longer confined to southern wilds; attempts were made to grow them in northern gardens, and valuable storage techniques were accumulated.

徐炫《五代新说》记述隋文帝时事：“蜀中人皆以蜡封蒂部，日久尤鲜。”
杜甫《树间》：“岑寂双柑树，婆娑一院香。”
(表明当时已有在庭院中种植柑橘的习惯)

Why Garden Cultivation and Tech Exploration?

Aesthetics Meet Utility

- Mandarin trees are evergreen, fragrant, and beautiful, aligning with the aesthetic tastes of Tang scholar-officials.
- The demands of the tribute system and the market directly spurred continuous experimentation and innovation in mandarin preservation and cultivation techniques.





The Practical Turn: Utility & Globalization (Song to Mordern)

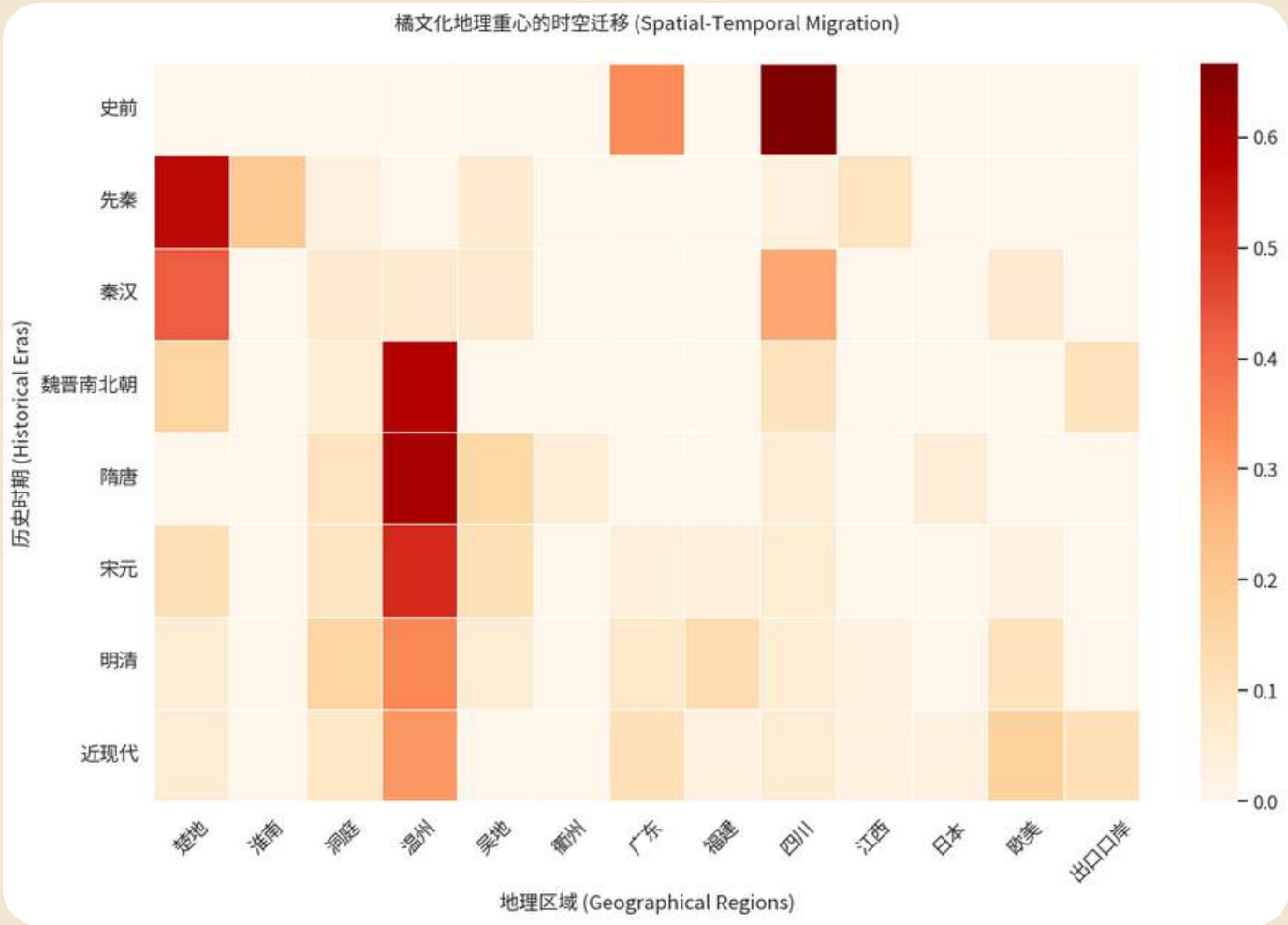
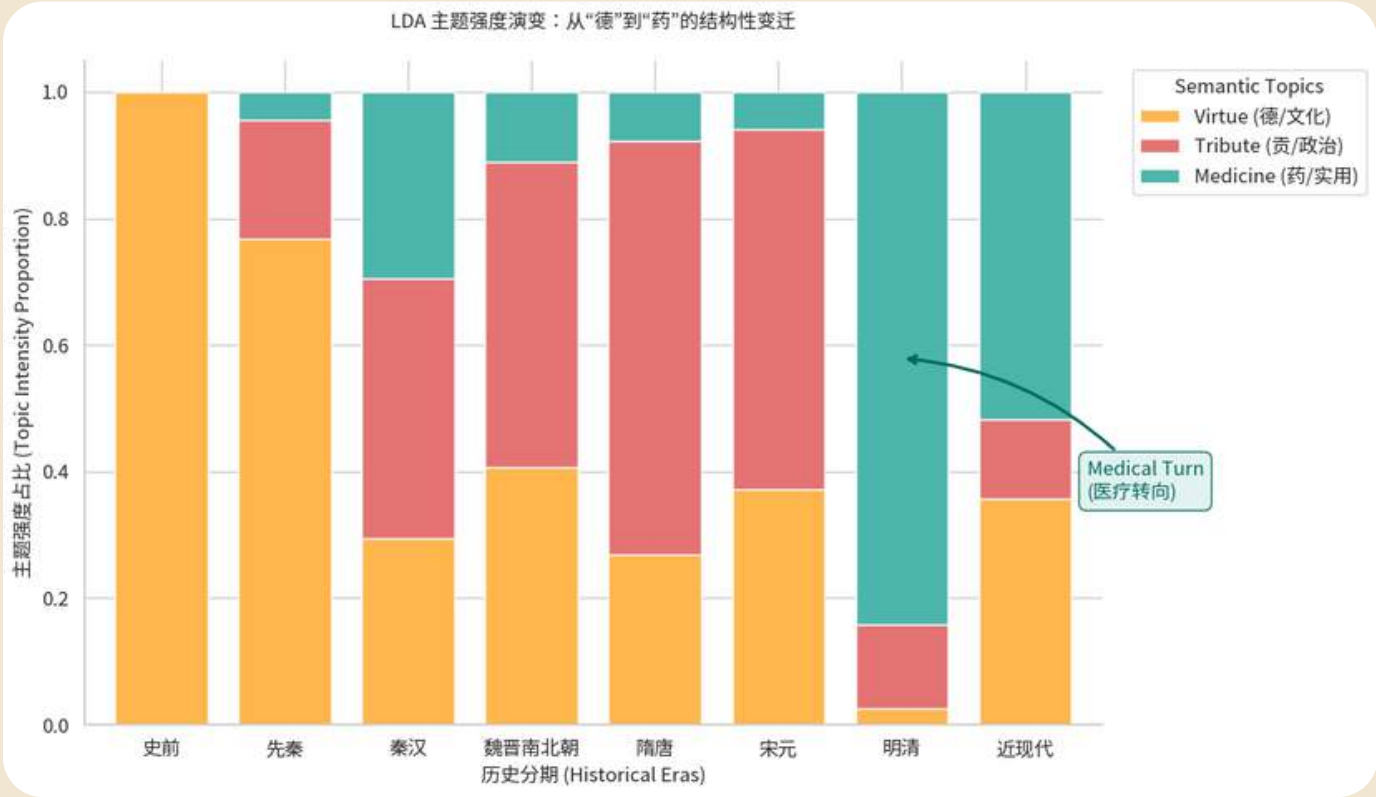
Era 3: 宋元明清 (Commodity & Medicine - The Chenpi System)

Subtitle 1: Medicinal & Culinary Fusion (药食同源)Value shift: From Fruit to Medicine

- Authoritative Endorsement: Li Shizhen's Compendium of Materia Medica (《本草纲目》) defined citrus's medicinal functions (peel for digestion, leaves for liver health), elevating it from mere fruit to a certified dual-use resource.
- Holistic Utilization: A mature model emerged—consuming the pulp while sun-drying the peel into Chenpi (陈皮) for medicine, perfectly embodying the "Medicinal & Culinary Fusion" symbol.

Subtitle 2: Commercial Reach (商脉致远)Symbolic upgrade: From Local Produce to Trade Commodity

- The "Long-Distance Gene":
- Empirical Trade Evidence



2. Historical Drivers of the Symbolic System

1: Technological Breakthrough (技术突破)

- Julu (《桔录》) → Democratized Cultivation: Systematized knowledge (planting, grafting, storage) broke the imperial monopoly.
- Result → Mass Production: Enabled output to surge, securing the material basis for both medicinal use and commercial trade.

2: Medicinal Empowerment (医药赋能)

- Compendium of Materia Medica (《本草纲目》) → Authoritative Endorsement: Li Shizhen's work legitimized the medicinal value of citrus, especially its peel (Chenpi) and leaves.
- Result → From Folk Remedy to Certified Medicine: Solidified the "Medicinal & Culinary Fusion" (药食同源) symbol, as confirmed by Word2Vec analysis linking citrus directly to "Li Shizhen."

3: Commercial Transformation (商业转型)

- Rise of Private Trade Networks (明清) → Broke Tribute System's Grip: Citrus's durability ("致远") enabled long-distance trade via canals and sea routes.
- Result → From Tribute to Commodity: Geographic data shows a shift to new production hubs (Fujian, Guangdong) and export ports, cementing its role as a key commodity and paving the way for globalization.



3. Value Sublimation: The Civilizational Code of Fusing Medicine & Commerce

Subtitle 1: Synergy of Technology & Medicine: Embodying Eastern Wisdom

Subtitle 2: A Microcosm of Commercial Evolution: From Monopoly to Global Integration

Subtitle 3: A Vehicle for Cultural Export: The Global Extension of Value





Era 4: Globalization

1.The Dual History of "Mandarin"

- **Botanically:** A class of citrus fruits (the **Mandarin Orange**).
- **Linguistically:** The official language of China (**Mandarin Chinese**)
- **The Core Inquiry:** How did one word come to represent both? What hidden historical logic and cultural narrative explains this seemingly coincidental link?



2. Historical Origins: Tracing the Symbolic Link

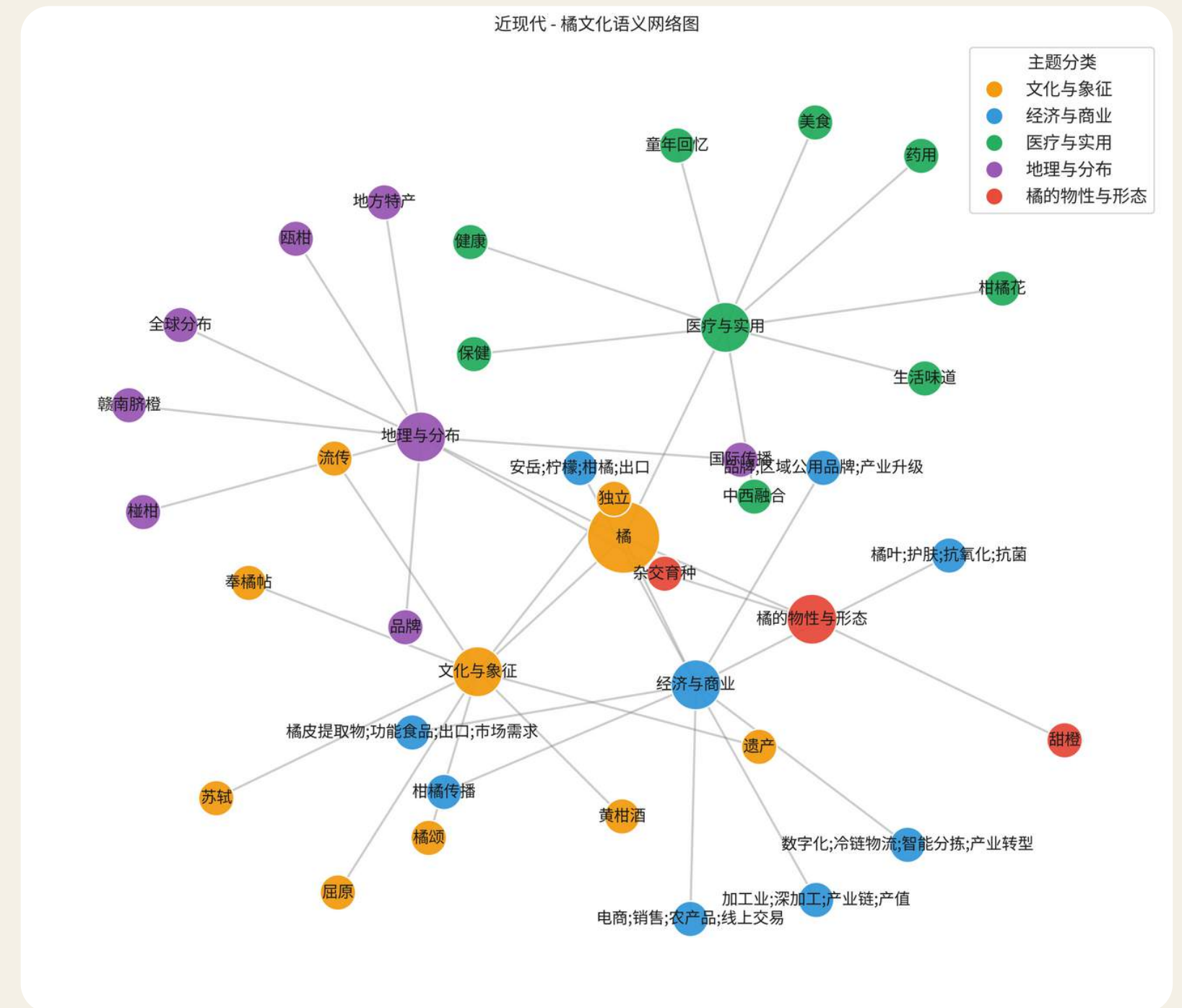
1. Phase I: An Eastern Product Becomes a Symbol (From 15th C.)

From the 15th century, the "Chinese Apple" (sweet orange) arrived in Europe as a symbol of exotic value from the East, establishing "China" as a byword for rarity and prestige in the age of discovery.

1. Phase II: Extending the Name: From Fruit to Language (19th C.)

By the 19th century, this association was solidified when the fruit was named "Mandarin" in America, directly borrowing the title for high-ranking Chinese officials to signify its superior and precious status.

Essentially, naming it "Mandarin" was an act of value projection, attributing to the fruit the qualities of an imperial official: a treasure that was prestigious, superior, and officially representative, thereby securing its special status in global horticulture.





3. Elevated Significance: From Micro-history to Grand Narrative

1.Semantic Trajectory: From a Concrete Symbol to an Abstract System

Explanation: This title uses "Semantic Trajectory" to highlight the word's clear developmental path. It pinpoints how the meaning evolved from a concrete symbol of "Chineseness" (the fruit) to a complex abstract system (the language).

2. A Global History Perspective: The Transnational Flow from Species to Culture

- (1) The migration of a species
- (2) The evolution of a word
- (3) The export of a culture

『橘』的语义演变轨迹 (3000年 Word2Vec 时代切片模型)

近现代

名称

源自

来源

珍贵

就是

“一个由身着华服的官员（Mandarins）说着（Mandarin Chinese）带着中华金橘（Mandarin Oranges）走向世界”

Academic Insights

Insight 1: The Process of Cultural Disenchantment

"The Descent from the Altar" Our multi-dimensional data jointly confirm a grand historical trend: oranges have undergone a 3,000-year-long process of "disenchantment".

- It originated from the "divinity" of the Pre-Qin period (the moral totem in Qu Yuan's works);
- During the Han and Tang dynasties, it transformed into "political" (auspicious signs of imperial power);
- Finally, during the Ming and Qing dynasties, it completely evolved towards "physical properties" (being disassembled into medicinal materials such as dried tangerine peel and tangerine pith).
- Conclusion: The history of oranges is a microcosm of Chinese culture's journey from elite idealism to secular pragmatism.

Insight 2: Symbolic Re-contextualization

Semantic network analysis of "Context is King" shows that the meaning of symbols is not static, but rather reconstructed as the "Contextual Container" changes.

- When the orange is placed in "Chu Ci", it represents a gentleman;
- When placed in "Ben Cao", it is a medicine;
- When placed in the "Customs Report", it is a commodity.
- Conclusion: The vitality of cultural symbols does not lie in clinging to their original meanings, but in their ability to continuously embed themselves in new social contexts (political, medical, commercial) to achieve rebirth.

Insight 3: The Value of Distant Reading

Traditional humanities research excels at "Close Reading" texts, while our AI model enables "Distant Reading".

- We have captured micro-semantic drift that is difficult to detect with the naked eye, such as the annual decline in the weight of the character "德" and the exponential increase in the weight of the character "药".
- Conclusion: This quantitative perspective provides objective, structured new evidence for our understanding of the evolution of Chinese culture.



Thank you for listening!

Any Questions?